Princy Ranaivomanana, PhD

Data Science & Research Expertise | ML Engineer | MLOps

Netherlands · Open to EU/International Roles

https://rtprincy.github.io rtprincy Google Scholar

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PROFILE

My name is Princy. I am originally from Madagascar. I have recently completed a PhD in Astronomy. I chose this topic as I love turning (observational) data into insights and improving our knowledge of the universe. I have experiences in analysing and processing large datasets (structured and unstructured), modelling data to uncover trends and patterns, and developing (un)supervised machine learning models, which I acquired during my PhD research. Besides, I am flexible, an independent and team player, and always curious and eager to learn new skills.

SKILLS

- Machine learning: Perform classification or clustering tasks on (un)structured data
- Deep Learning: Perform synthetic data/image generation tasks using generative AI models
- Python: Conduct exploratory data analysis with libraries, such as Pandas, NumPy, and Scikit-learn
- SQL: Perform SQL query to retrieve and manipulate large datasets from astronomical databases

COMPETENCIES

- Problem-solving
- Technical communication (reporting and presentations)
- Mentoring and supervision

PROFESSIONAL EXPERIENCE

1. Industry Collaboration Project – Nutrico *16-18 Feb 2026*

Annual modelling camp organised by the Mathematics Department, Radboud University.

Collaborated within a multidisciplinary team of researchers to develop machine learning solutions for genotype determination at the Nutrico (Trouw Nutrition) company, reducing processing time from one week to several minutes.

2. PhD Researcher (Astronomy) *Radboud University & KU Leuven | Oct 2021 – Sep 2025*

a) Project leader for a machine learning effort involving the classification of massive time-series data.

- Outcome: Distinguished various clusters of objects using t-SNE/UMAP and Gaussian mixture models.
- Deliverable: Two peer-reviewed publications in a high-index journal.

b) Contributed to the solution of periodicity search in irregularly sparsely sampled time series.

c) Authored four peer-reviewed publications.

3. **Teaching Assistant** *Radboud University & KU Leuven | Oct 2021 – Sep 2025*

- Tutored courses on data analysis and astronomical instrumentation
- Mentored two bachelor research projects (two students per project)

EDUCATION

PhD, Astronomy & Astrophysics (Dual Degree): Radboud University & KU Leuven | 2021 – 2025

MPhil, Astronomy: University of Manchester, United Kingdom | 2020 – 2021

MSc, Astronomy: University of Antananarivo, Madagascar | 2016 – 2017

BSc, Physics: University of Antananarivo, Madagascar | 2012 – 2015

PRIZES AND GRANTS

- Receipt of the LKBF grants for attending conferences (2023/2024/2025)
- Recipient of the UK's Newton Fund/DARA Big Data Masters Bursary (2020)
- DARA travel grant − African Radio Interferometry Winter School − South Africa (2019)
- Receipt of SKA/SARAO grant to attend the 2nd Big Data Africa School − South Africa (2018)
- IAU travel grant − The 39th International School for Young Astronomers (ISYA) − Ethiopia (2017)

RELEVANT CONFERENCES ATTENDED & CONTRIBUTIONS

Invited speaker:

- Invited talk at the Hamburg Observatory Colloquium (November 2025)
- Talk title: **Unsupervised machine learning** for variability analysis in time-domain astronomy

Conference talk:

- Conference: The 12th international meeting on hot subdwarfs & related objects (May 2025)
- Talk title: Unravelling Hot Subdwarfs and Binaries Variability with Gaia DR3 and **Machine Learning**

RELEVANT CERTIFICATION

- Oxford machine learning school 2025 (issued by AI for Global Goals) Online Certificate
- Oxford machine learning school 2023 (issued by AI for Global Goals) Online Certificate
- Advanced Machine Learning and Signal Processing (issued by Coursera) Online Certificate
- Data Analysis with Python (issued by Coursera) Online Certificate
- Exploratory Data Analysis for Machine Learning (issued by Coursera) Online Certificate

SELECTED STRENGTHS FOR INDUSTRY

- Translating complex problems into understandable solutions
- Quick and flexible learner
- Working with imperfect, real-world data at scale